
EMCH 574 – Vibration and Waves

Homework 03

Student J.C. Vaught

Due date TBD

Contents

1	String vibration	3
1.1	A.1 General Analysis Of String Vibration	3
1.2	A.2 Analyze Natural Frequencies And Modeshapes Of A Vibrating String	5
1.3	A.3 Calculate String Vibration	7
1.4	A.4 Calculate D’Addario Guitar Strings	8
1.5	A.5 Calculate Vibration Of A Plucked String	10
1.6	A.6 Calculate Frequency Response Of A Struck String	11
2	Flexural vibration	13
2.1	B.1 General Analysis Of Flexural Vibration	13
2.2	B.2 Analyze Cantilever Beam Vibration	15
2.3	B.3 Calculate Cantilever Beam Vibration	17
2.4	B.4 Calculate The 512 C Tuning Fork	18
2.5	B.5 Calculate Vibration Of A Plucked Reed	20
3	Extra Credit Challenge Problems	21
3.1	C.1 Analyze Flexural Vibration In A Free-Free Beam	21
3.2	C.2 Calculate Wind-Chime Tones	22
3.3	C.3 Calculate Flexural Vibration In A Free-Free Beam	24
3.4	C.4 Analyze Flexural Vibration Of Simply Supported	25
3.5	C.5 Calculate Simply Supported Flexural Vibration	26
3.6	C.6 Analysis Of Wave Equation In A Slender Bar	27
3.7	C.7 Calculate Axial Vibration With Various Boundary Conditions	28
3.8	C.8 Calculate The Hopkinson Bar	29
3.9	C.9 Analyze Axial Vibration In A Free-Free Bar	30
3.10	C.10 Analyze Axial Vibration In A Fixed-Fixed Bar	32
3.11	C.11 Analyze Axial Vibration In A Fixed-Free Bar	33

Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items
- UG students are not required to do the G work, but they may do it for bonus points
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement [reference](#)
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga...) [reference](#)
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit
- Normalized plots are plots with the [y-axis](#) scaled between -1 and +1. A plotted variable is normalized by dividing it by its maximum absolute value

Notations and Abbreviations

- BVP = boundary value problem
- EOM = equation of motion
- FBD = free body diagram
- FRF = frequency response function
- IVP = initial value problem
- LHS = left hand side
- NLM = Newton's law of motion
- ODE = ordinary differential equation
- PDE = partial differential equation
- RHS = right hand side

1 String vibration

A.1 General Analysis Of String Vibration

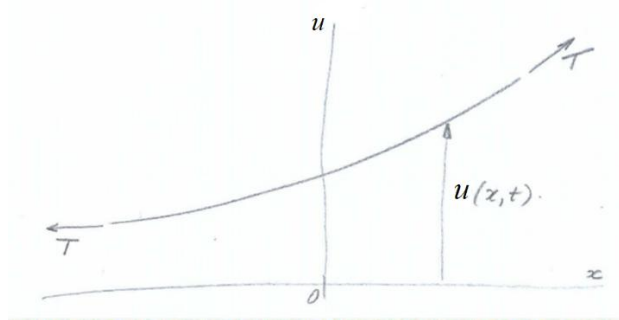


Figure 1 Taut string under tensile load T : (a) general view; (b) infinitesimal element

Consider a taut string of mass m per unit length stretched by a constant tension load T (Figure 1). The string undergoes small-amplitude transverse vibration with frequency ω as described by the following formula: $u(x,t) = U(x)e^{-i\omega t}$ (vibration motion) (1) Do the following:

- (1) state the assumption made about the string tension T during the vibration motion u
- (2) draw free-body diagram (FBD) of an infinitesimal string element of length ds and write sum of y forces in terms of string tension T and slope α
- (3) apply small-displacement approximation and write the approximate expressions for ds , $\sin \alpha$, $\sin(\alpha + d\alpha)$
- (4) write the linearized expression for the sum of y forces
- (5) express the slope increment $d\alpha$ in terms of space increment dx and the relevant partial derivative of the displacement $u(x,t)$
- (6) apply Newton's law of motion (NLM) and write the NLM equation
- (7) write the governing equation for string vibration in terms of tension T , mass per unit length m , and the relevant partial derivatives of the displacement $u(x,t)$
- (8) define wavespeed c for wave traveling in a taut string in terms of tension T and mass per unit length m
- (9) write the wave equation
- (10) substitute Eq. (1) into item (9) and eliminate the time dependency $e^{-i\omega t}$ to obtain an ODE defining $U(x)$
- (11) recall the definition of the wavenumber γ and substitute into (10) to obtain an ODE in terms of only γ and $U(x)$
- (12) assume exponential solution of the form $U(x) = e^{sx}$
- (13) develop the characteristic equation for s
- (14) solve the characteristic equation and write the roots s_1 , s_2
- (15) substitute (14) into (12) and write the general solution for $U(x)$ in terms of constants A_1 , A_2

and complex exponentials depending on γ

- (16) use Euler's formulae to rewrite (15) in terms of new constants C_1 , C_2 and trigonometric functions depending on γ
- (17) substitute (16) into Eq. (1) and write the general solution for $u(x, t)$ in terms of constants C_1 , C_2 , trigonometric functions of γx and a complex exponential function of ωt
- (18) Graduate Work: write the general solution for $u(x, t)$ in terms of complex exponential functions of γ and ω

A.2 Analyze Natural Frequencies And Modeshapes Of A Vibrating String



Figure 2 Taut string of length L under constant tension T

Consider a taut string of length L and mass m per unit length. The string is pinned at both ends and stretched by a constant tension load T (Figure 2). The string undergoes transverse vibration with frequency ω . Do the following:

- (1) write the boundary conditions at the two ends of string
- (2) recall from Section A.1 the general solution for $u(x,t)$ expressed the multiplication of two terms:
 - (a) first term: a function $U(x)$ depending on constants C_1 , C_2 and trigonometric functions of γx
 - (b) second term: a complex exponential function of ωt
- (3) substitute (2) into (1) and write boundary conditions in terms of constants C_1 , C_2
- (4) solve (3) for C_2 and obtain an equation for C_1
- (5) discuss the trivial solution
- (6) choose to search for a nontrivial solution and define the equation that needs to be solved to find a nontrivial solution
- (7) introduce the notation $z = \gamma L$ and write the eigenvalue equation in terms of z
- (8) solve the eigenvalue equation and write the general expression for the eigenvalues z_j , $j = 1, 2, 3, \dots$
- (9) use (8) to write the general expression for the wavenumber γ_j in terms of j and length L
- (10) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c
- (11) solve (10) for frequency ω in terms of wavenumber γ and wavespeed c
- (12) use (9) and (11) to write the general expression of the natural frequency ω_j in terms of j , wavespeed c , and string length L
- (13) use (12) to write the general expression of the Hz frequency f_j in terms of j , wavespeed c , and string length L
- (14) explain the meaning of fundamental frequency and overtones
- (15) recall from HW01 the relationship between wavelength λ and wavenumber γ
- (16) use (9) into (15) to write the general expression for the wavelength λ_j in terms of j and string length L
- (17) use (16) to explain which wavelength multiples could accommodate vibratory standing waves in a string of length L fixed at both ends

- (18) recall from (2)(a) the expression for $U(x)$
- (19) use (4) to eliminate C_2
- (20) set $C_1 = 1$ in (19) and write the general expression of the modeshape $U_j(x)$ in terms of the wavenumber γ_j
- (21) give the expression of modeshape orthonormality

A.3 Calculate String Vibration

Consider a taught string of length $L = 22\text{ in}$ tuned to the note G3. The purpose is to calculate the string vibration characteristics and the modeshapes for $j = 1, \dots, N$ modes with $N = 5$. We should also verify mode orthonormality using numerical integration on a `linspace(0, L, Nx)` with $N_x = 1000$. Recall the appropriate results from in-class instruction class and BB notes to do the following:

- (1) Display input data
- (2) State the note symbol (letter and octave) and the corresponding fundamental Hz frequency f_1 of the string.
- (3) Use frequency f_1 and length L to calculate the wavespeed c
- (4) Calculate and display modal wavenumbers γ_j and frequencies ω_j
- (5) Calculate the modeshapes, plot each mode separately and then plot overlapped all the N modes in a single plot
- (6) Verify modeshape orthonormality by numerical integration
- (7) Discuss your results

A.4 Calculate D'Addario Guitar Strings

NORMAL 6-STRING SET					
E	A	D	G	B	E
6 th	5 th	4 th	3 rd	2 nd	1 st
					
—	—	—	—	—	—
BRASS	RED	BLACK	GREEN	PURPLE	SILVER

Table 1

Brass Red Black Green Purple Silver

musical

E A D G B E

note *M*, grams 6.42 4.18 2.45 1.332 0.655 0.367 *T*, lbf 24.95 28.93 29.93 30.05 23.31 23.36

Consider a set of six light-gauge d'Addario guitar strings. All six strings have the same length $L = 25.0$ in. The mass M and the tension T of each string are given in in Table 1.

- (1) Display input data For each of the six strings, do the following:
- (2) Display string length L in m
- (3) Calculate and display distributed mass, m , in kg/m
- (4) Display tension T in N
- (5) Calculate wavespeed c in m/s
- (6) Calculate fundamental frequency f_1 in Hz
- (7) Find the musical note that most closely approximates the frequency and give its letter name and octave number (e.g., G3 signifies note G in the 3rd octave)
- (8) Display the frequency of the musical notes found at (7)
- (9) Calculate the percentage difference between your findings and the nominal frequency of that musical note
- (10) Comment on your findings and identify possible sources of error.
- (11) Graduate work: Calculate the necessary adjustments in mass, tension, or both to achieve

frequency match within 0.00%

A.5 Calculate Vibration Of A Plucked String

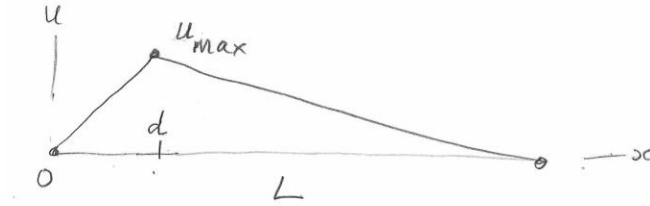


Figure 3 Taut string plucked at an arbitrary location d

Consider a taut string of length $L = 25.5$ in tuned to the note G3. The string is plucked by applying an initial displacement $u_{max} = 1.25$ mm at a location d . In this problem, we will consider three possible plucking locations: $d = L/2$, $L/4$, $7L/10$. The purpose is to calculate the response $u(x, t)$ of the plucked string using the modal expansion method with $N = 10$. Recall the appropriate results from in-class instruction class and BB notes to do the following:

- (1) Display input data
- (2) State the note symbol (letter and octave) and the corresponding fundamental Hz frequency f_1 of the string. Calculate the corresponding fundamental mode period τ_1 .
- (3) Use frequency f_1 and length L to calculate the wavespeed c
- (4) Calculate modal wavenumbers γ_j and frequencies ω_j
- (5) Calculate and plot overlapped the N modes
- (6) For each plucking location, do the following:
 - (a) display the value d and plot the initial shape of the string $u_0(x)$
 - (b) calculate by numerical integration the modal participation factors A_j , display A_j in mm, and do a bar plot
 - (c) plot the time response of the string at the plucking location $u(d, t)$ for $t = 0, \dots, t_{max}$ with $t_{max} = 3\tau_1$
 - (d) Graduate work: reconstruct the initial string shape and plot it overlapped on the actual shape
- (7) Discuss comparatively the results for the three plucking locations

A.6 Calculate Frequency Response Of A Struck String

This problem addresses the frequency response of a taut string when struck by a point load like in the case of the hammered dulcimer (Figure 4).

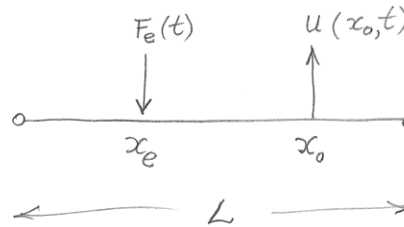


Figure 4 Struck string: (a) the hammered dulcimer; (b) schematic

Consider a taut string of length $L = 25.5$ in and distributed mass $m = 0.0015$ kg/m tuned to the note G3. The string damping ratio is assumed to have same value $\zeta_0 = 2.1\%$ in all modes. The string is struck at location x_e . The response of the string is measured with a piezoelectric acoustic pickup at location x_0 . Three x_e, x_0 cases are considered: case x_e, x_0

- (a) $L/3, L/3$
- (b) $L/3, L/2$
- (c) $L/2, L/2$ The striking force is assumed harmonic with amplitude $F_e = 0.5$ N and frequency range $f_{\text{start}} = 20$ Hz through $f_{\text{end}} = 2000$ Hz with $N_f = 5000$ steps. The objective is to calculate and plot the frequency response

function (FRF) over the frequency range using modal expansion method with $N = 10$ modes. Do the following:

- (1) Display input data
- (2) State the note name and octave and the corresponding fundamental Hz frequency f_1 of the string. Calculate the corresponding period τ_1 and the wavespeed c
- (3) For $j = 1, \dots, N$, calculate and display the modal properties wavenumber γ_j , frequency ω_j , mass m_j ; damping ζ_j and plot overlapped the modeshapes
- (4) Define the frequency range f using `linspace( $f_{\text{start}}, f_{\text{end}}, N_f$ )` and then define $\omega = 2\pi f$
- (5) For each x_e, x_0 case, plot the $|\text{FRF}(\omega; x_e, x_0)|$ in dB and find the frequencies and nearest musical notes, as well as the index of the string modal frequency
- (6) Graduate work: plot the time response of the normalized sound picked up at $x_0 = L/2$

when the string was struck at $x_e = L/3$. The excitation frequency should be the dominant note frequency in the FRF plot. The time plot should cover three periods of the dominant frequency.

2 Flexural vibration

B.1 General Analysis Of Flexural Vibration

Consider a uniform beam of mass m per unit length and bending stiffness EI undergoing flexural motion with vertical displacement $w(x, t)$ as shown in Figure 5. Centroidal axes are assumed.

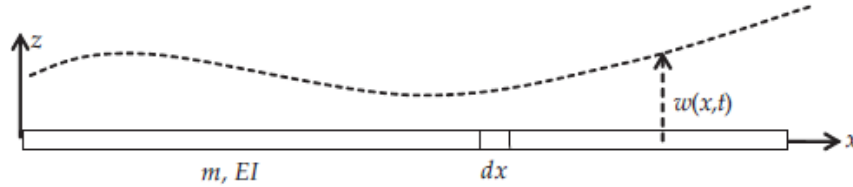


Figure 5 Beam undergoing flexural vibration

Do the following:

- (1) state the assumptions needed for performing the flexural vibration analysis
- (2) draw free-body diagram (FBD) of an infinitesimal beam element dx and show the stress resultants N , M , V .
- (3) write the definitions of the stress resultants N and M
- (4) use stress-strain relation to write the stress resultants in terms of strains
- (5) perform kinematic analysis to express the stress resultant M in terms of partial derivatives of the displacement w
- (6) perform free-body analysis of the infinitesimal element dx and find the sum of forces ΣF_z and sum of moments ΣM
- (7) apply Newton's law of motion (NLM) in the z direction and obtain an equation for the shear force V
- (8) apply NLM in the rotational direction and obtain an equation for the moment M
- (9) substitute (6) into (8) and get an equation that relates M and V
- (10) use (5) into (9) and obtain the governing equation
- (11) define the flexural constant a in terms of E , I , m
- (12) use (11) into (10) to obtain the governing equation as a PDE in x , t , and flexural constant a
- (13) assume harmonic vibration of the form $w(x, t) = W(x)e^{-i\omega t}$
- (14) substitute (13) into (12) and obtain an ODE in x with constants a , ω
- (15) define the wavenumber γ in terms of ω and a and substitute in (14) to write the ODE with only one constant, γ
- (16) assume $W(x)$ as an exponential function of x of the form $= e^{sx}$
- (17) substitute (16) into (15) and derive the characteristic equation
- (18) find the four roots of the characteristic equation

- (19) write the general solution $W(x)$ in terms of constants A_1, A_2, A_3, A_4 and complex exponential functions of γx
- (20) rewrite the general solution (19)(13) in terms of constants C_1, C_2, C_3, C_4 multiplying trigonometric and hyperbolic functions of γx
- (21) Graduate Work: explain how one can find the constants $C_1,$

C_2, C_3, C_4

B.2 Analyze Cantilever Beam Vibration

A cantilever beam is fixed at one end and free at the other end. Consider a cantilever beam of length L as shown in Figure 6. The beam has mass density per unit length m , elastic modulus E , and cross-section second moment of area I .

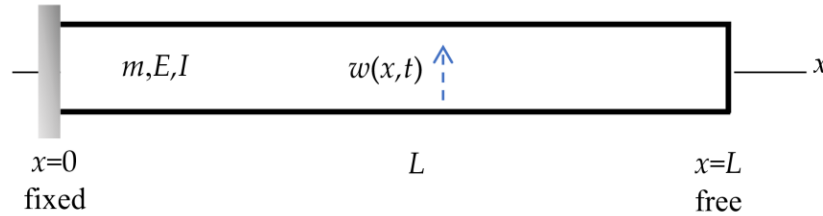


Figure 6 Vibration of a cantilever beam of length L . The cantilever beam is fixed at one end and free at the other end

Do the following:

- (1) write the boundary conditions at the two ends of the bar
- (2) recall from the notes the general solution $w(x,t)$ consisting of the multiplication of two terms:
 - (a) first term: a function $W(x)$ depending on constants C_1, C_2, C_3, C_4 , and trigonometric and hyperbolic functions of γx
 - (b) second term: a complex exponential function of ωt
- (3) substitute (2) into (1) and write boundary conditions in terms of constants C_1, C_2, C_3, C_4
- (4) solve (3) for C_3, C_4 in terms of C_1, C_2 and obtain a set of equations for C_1, C_2
- (5) discuss the trivial solution
- (6) choose to search for a nontrivial solution and define the equation that needs to be solved to find a nontrivial solution
- (7) introduce the notation $z = \gamma L$ and write the transcendental eigenvalue equation in terms of z
- (8) use the MATLAB function fzero to solve the transcendental eigenvalue equation and find $z_j, j = 1, \dots, N$ with $N = 5$
- (9) use item (7) to write the general expression for the wavenumber γ_j in terms of z_j and length L
- (10) Recall from Problem B.1 the definition of wavenumber γ in terms of frequency ω and flexural constant a
- (11) solve (10) for frequency ω in terms of wavenumber γ and flexural constant a
- (12) use (9) and (11) to write the general expression of the natural frequency ω_j in terms of eigenvalue z_j , beam constant a , and beam length L
- (13) recall definition of a in terms of bending stiffness EI and mass per unit length m and obtain an expression for the natural frequencies ω_j and f_j in terms of z_j, EI, m, L
- (14) use (4) to obtain an expression for the ratio C_1/C_2 in terms of wavenumber γ and beam length L . Denote this expression to be $-\beta$
- (15) recall from (2)(a) the expression for $W(x)$

- (16) use (4) and (14) to write the constants C_1 , C_3 , C_4 in terms of C_2
- (17) substitute (16) into (15) and write $W(x)$ in terms of C_2 , γx , β
- (18) Set $C_2 = 1$ and write the modeshapes $W_j(x)$ in terms of γ_j and β_j
- (19) give the expression of modeshape orthonormality
- (20) Graduate work:
 - (a) write the approximate expression for the eigenvalues z_j with $j \geq 6$
 - (b) solve the transcendental equation at item (7) for $N = 10$ and compare with the prediction using the approximate expression of item (a)

B.3 Calculate Cantilever Beam Vibration

Assume a cantilever beam of length $L = 445 \text{ mm}$ and cross-section dimensions $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The material is steel with $E = 200 \text{ GPa}$ and $\rho = 7850 \text{ kg/m}^3$. Do the following:

- (1) display input data
- (2) calculate the cross-section area A
- (3) calculate mass per unit length m
- (4) calculate second moment of area I
- (5) calculate flexural stiffness EI
- (6) calculate flexural constant a
- (7) solve the transcendental equation defined in Problem B.2 item (7) to find z_j , $j = 1, \dots, N$ with $N = 5$
- (8) calculate the wavenumbers γ_j
- (9) calculate angular frequencies ω_j
- (10) calculate Hz frequencies f_j
- (11) calculate the modeshape coefficients β_j
- (12) calculate the modeshapes $W_j(x)$, plot each mode separately and then plot overlapped all the N modes on a single plot
- (13) verify modeshape orthonormality by numerical integration
- (14) comment on your results

B.4 Calculate The 512 C Tuning Fork

Consider the 512 C aluminum-alloy tuning fork (Figure 7). Common values for aluminum elastic modulus and density are $E = 70 \text{ GPa}$ and $\rho = 2700 \text{ kg/m}^3$. This tuning fork is stated to give the middle C note (C_5) according to the scientific pitch scale reference. In the scientific pitch scale, the middle C note C_5 has the frequency $f_1 = 512 \text{ Hz}$.

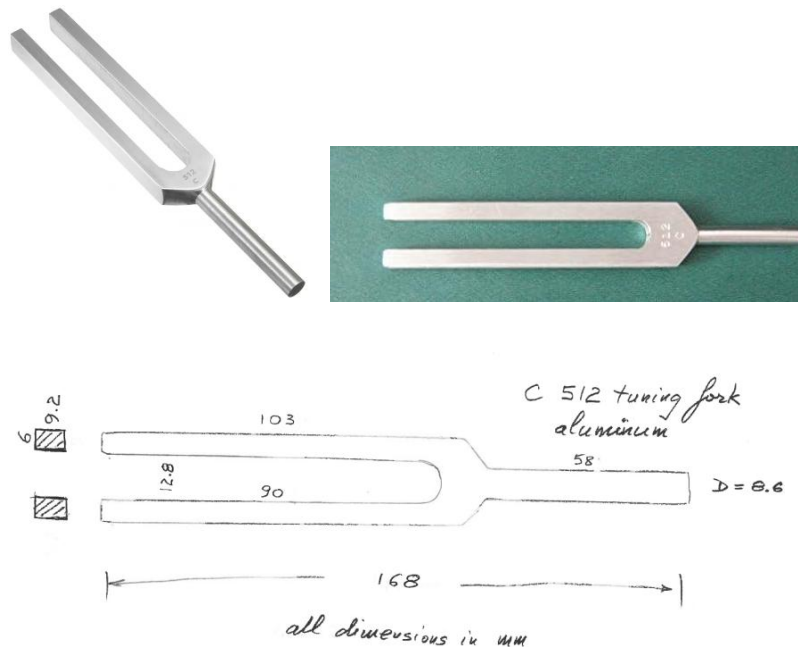


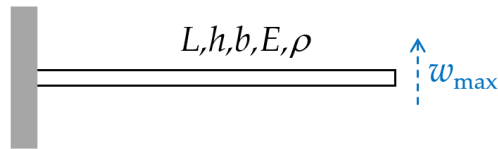
Figure 7 512 C tuning fork reference Tuning-Fork/dp/B00BLBE8QE . The sketch is done by DrG by measuring the actual fork.

Use the cantilever-beam analysis developed in Problem B.2 to perform the engineering evaluation of this tuning fork. Do the following:

- (1) display input data
- (2) use the sketch in Figure 7 to estimate the cantilever-beam length L . Hint: use engineering intuition to estimate where the root of the cantilever beam could be assumed
- (3) use the sketch in Figure 7 to obtain the cantilever-beam cross-section dimensions h, b
- (4) calculate cross-section area A
- (5) calculate mass per unit length m
- (6) calculate second moment of area I
- (7) calculate flexural stiffness EI
- (8) calculate flexural constant a
- (9) use the results of Problem B.3 to calculate the fundamental frequency f_1 in Hz
- (10) compare the calculated f_1 with the nominal tuning fork frequency of 512 Hz, calculate the percentage difference, and comment on your results

- (11) manually adjust the estimated beam length to a new value L_{new} until the new frequency f_{new} equals 512 Hz
- (12) comment on the engineering estimation and adjustment process developed in the solution of this problem
- (13) Graduate work: find the frequencies of the next three higher harmonics of the tuning fork after length adjustment

B.5 Calculate Vibration Of A Plucked Reed



Consider a cantilever beam of length $L = 14 \text{ mm}$ and cross-section dimensions $h = 0.3 \text{ mm}$, $b = 0.75 \text{ mm}$. The material is steel with $E = 200 \text{ GPa}$ and $\rho = 7850 \text{ kg/m}^3$. The beam is plucked with an initial tip displacement $w_{\max} = 0.15 \text{ mm}$. The purpose is to calculate the response $w(x, t)$ of the plucked beam using the modal expansion method. Recall the appropriate results from class and/or BB notes and do the following:

- (1) Display input data
- (2) Calculate cross-section area A , mass per unit length m , cross-section second moment of area I , flexural stiffness EI , and flexural constant a
- (3) Calculate the roots of the cantilever eigenvalue equation for $j = 1, \dots, N$ with $N = 5$
- (4) Calculate the fundamental Hz frequency f_1 and state the symbol (letter and octave) of the nearest musical note. Calculate the fundamental mode period τ_1
- (5) Recall the formulas for cantilever beam modeshapes and plot the modes $j = 1, \dots, N$
- (6) calculate and plot the initial deformed shape of the beam $w_0(x)$
- (7) Assume the modal expansion method, calculate the modal coefficients A_j , $j = 1, \dots, N$, and do a bar plot
- (8) plot the time response of the tip of the beam $w(L, t)$ for three fundamental-mode periods
- (9) Comment on your results

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Flexural Vibration In A Free-Free Beam

- (1) sketch a free-free beam indicating all the information needed for the analysis
- (2) write the boundary conditions at the two ends of the beam in terms of displacement $w(x, t)$ and stress resultants $M(x, t)$, $V(x, t)$
- (3) repeat all the steps of Section B.2

C.2 Calculate Wind-Chime Tones

Consider the wind chime shown in Figure 8 and Figure 9.



Figure 8 Wind chime

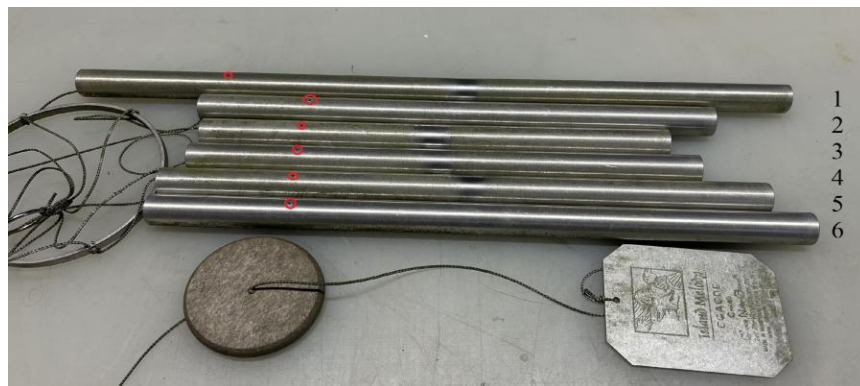


Figure 9 Tubes of the wind chime with the suspension points circled in red.

The wind chime tubes are made of aluminum with $E = 70 \text{ GPa}$ and $\rho = 2700 \text{ kg/m}^3$. The outer and inner diameters of the tubes are $D = 22.3 \text{ mm}$; $d = 19.4 \text{ mm}$. The tube length is L . Each tube is suspended at a distance e from its end. The L and e values are given in Table 2.

Table 2: The measured L and e values of the six wind chime tubes # L , mm e , mm

1	500	110
2	352	78
3	312	70
4	332	74
5	384	87
6	406	90

Use the free-free beam analysis developed in Problem C.1 to perform the engineering evaluation of the wind chime tubes. Do the following:

- (1) calculate the cross-section area A

- (2) calculate mass per unit length m
- (3) calculate second moment of area I
- (4) calculate flexural stiffness EI
- (5) calculate flexural constant a
- (6) use the MATLAB function `fzero` to find the first root of the transcendental eigenvalue equation defined in Problem C.1
- (7) use the length values given in Table 2 to calculate the wavenumber γ and the corresponding frequencies ω_1 and f_1 for each of the six tubes
- (8) use [reference](#) to estimate the musical notes that might be associated with the frequencies of the six tubes
- (9) Graduate work:
 - (a) calculate and plot the modeshape $W_1(x)$ for each of the six tubes
 - (b) estimate the location x_0 where the modeshape $W_1(x)$ crosses the zero line, compare the values x_0 with the values e given in Table 2, and calculate percentage difference. Comment on your results
- (10) Graduate work: find by manual iterations the adjusted tube length values that will give exact musical notes without # at item (8) above

C.3 Calculate Flexural Vibration In A Free-Free Beam

Assume a free-free beam of length $L = 445 \text{ mm}$, and cross-section dimensions $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The material is steel with $E = 200 \text{ GPa}$ and $\rho = 7850 \text{ kg/m}^3$. Do the following:

Repeat the steps of Section B.3

C.4 Analyze Flexural Vibration Of Simply Supported

BEAM

- (1) sketch a simply supported beam indicating all the information needed for the analysis
- (2) write the boundary conditions at the two ends of the beam in terms of displacement $w(x, t)$ and stress resultants $M(x, t)$, $V(x, t)$
- (3) recall from class notes the general solution $w(x, t)$ consisting of the multiplication of two terms:
 - (a) first term: a function $W(x)$ depending on constants C_1 , C_2 , C_3 , C_4 , and trigonometric and hyperbolic functions of γx
 - (b) second term: a complex exponential function of ωt
- (4) substitute (3) into (2) and write boundary conditions in terms of constants C_1 , C_2 , C_3 , C_4
- (5) solve (4) for C_2 , C_4 and obtain a set of equations for C_1 , C_3
- (6) discuss the trivial solution
- (7) choose to search for the nontrivial solution and define the equation that needs to be solved to find a nontrivial solution
- (8) introduce the notation $z = \gamma L$ and write the eigenvalue equation in terms of z
- (9) write the general expression of the roots z_j of the eigenvalue equation
- (10) use item (8) to write the general expression for the wavenumber γ_j in terms of j and length L
- (11) Recall from class notes the definition of wavenumber γ in terms of frequency ω and flexural constant a
- (12) solve (11) for frequency ω in terms of wavenumber γ and flexural constant a
- (13) use (10) and (12) to write the general expression of the natural frequency ω_j in terms of j , beam constant a , and beam length L
- (14) recall definition of a in terms of bending stiffness EI and mass per unit length m and obtain an expression for the natural frequencies ω_j and f_j in terms of j , EI , m , L
- (15) use item (5) to solve for C_3
- (16) substitute the values found so far for C_2 , C_3 , C_4 and write $W(x)$ in terms of C_1 and γx
- (17) Set $C_1 = 1$ and write the modeshapes $W_j(x)$ in terms of γ_j
- (18) give the expression of modeshape orthonormality

C.5 Calculate Simply Supported Flexural Vibration

Assume a simply supported beam of length $L = 445 \text{ mm}$, and cross-section dimensions $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The material is steel with $E = 200 \text{ GPa}$ and $\rho = 7850 \text{ kg/m}^3$. Do the following: Repeat the steps of Section B.3

C.6 Analysis Of Wave Equation In A Slender Bar

Assume a uniform slender bar of cross section A and density ρ (Figure 10). The bar undergoes time-varying axial displacement $u(x, t)$. The displacement is assumed uniform across the bar cross section.

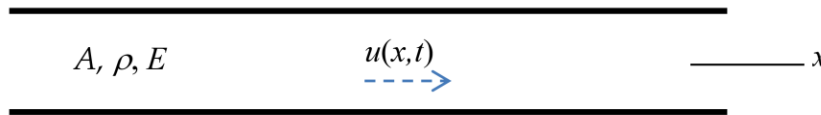


Figure 10 Wave motion in a uniform slender bar

Do the following:

- (1) state the assumption made about motion and deformation in the slender bar
- (2) draw free-body diagram (FBD) of an infinitesimal bar element of length dx and write Newton's law of motion (NLM) equation
- (3) write elasticity equation that relates stress and displacement
- (4) substitute (3) into (2) and derive the partial differential equation (PDE) in terms of displacement
- (5) define the wavespeed c in terms of elasticity modulus E and density ρ
- (6) write the wave equation
- (7) assume separation of x and t variables and write the solution as a product between an x function $U(x)$ and an exponential function in t of the form $e^{-i\omega t}$
- (8) substitute (7) into (6) and derive an ordinary differential equation (ODE) in terms of c , ω , $U(x)$
- (9) recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c and use it to simplify (8)
- (10) assume $U(x)$ as an exponential function of x of the form e^{sx}
- (11) substitute (10) into (9) and derive the characteristic equation
- (12) find the roots s_1 , s_2 of the characteristic equation in terms of the wavenumber γ and write the general solution $U(x)$ in terms of complex exponentials of γx
- (13) add the time dependency defined at (7) and write the general solution for $u(x, t)$ in terms of complex exponentials of γx , ωt , and constants A_1 , A_2
- (14) rewrite the general solution (13) in terms of constants C_1 , C_2 , trigonometric functions of γx and a complex exponential function of ωt
- (15) Graduate Work: identify the forward and backward waves in the general solution derived at (13)

C.7 Calculate Axial Vibration With Various Boundary Conditions

Consider a slender aluminum bar of diameter $D = 20 \text{ mm}$ and length $L = 3.0 \text{ m}$. Common values for aluminum elastic modulus and density are $E = 70 \text{ GPa}$ and $\rho = 2700 \text{ kg/m}^3$. The bar undergoes axial vibration with frequency ω . The purpose is to calculate the string vibration characteristics, the modeshapes, for $j = 1, \dots, N$ modes with $N = 5$. We should also verify mode orthonormality using numerical integration on a `linspace(0, L, Nx)` with $N_x = 1000$. Three boundary condition cases are considered:

boundary conditions case $x = 0$ $x = L$

- (a) fixed fixed
 - (b) fixed free
 - (c) free free
- (1) Display input data
 - (2) Calculate wavespeed c Then, for each of the three cases, do the following:
 - (3) Display boundary conditions
 - (4) Calculate and display modal vibration characteristics
 - (5) Calculate and plot overlapped the displacement modeshapes
 - (6) Verify modeshape orthonormality by numerical integration
 - (7) Calculate and plot overlapped the normalized stress modeshapes
 - (8) Discuss your results

C.8 Calculate The Hopkinson Bar



reference

The Hopkinson pressure bar is used to create and measure axial stress pulses. It consists of a long and slender metallic bar which is hit at one end. The hit creates a stress pulse that propagates in the bar and is measured by strain gauges. A common application is to measure the compression behavior of a small specimen placed in what is called the Split-Hopkinson bar. Consider an aluminum Hopkinson pressure bar of diameter $D = 20 \text{ mm}$. Common values for aluminum elastic modulus and density are $E = 70 \text{ GPa}$ and $\rho = 2700 \text{ kg/m}^3$. Do the following:

- (1) Calculate the axial wavespeed c in aluminum
- (2) Assume the bar length is $L = 3.2 \text{ m}$ and calculate the time t required to travel the length of the bar
- (3) Calculate the natural frequencies f_1, f_2, f_3 of the first three modes of bar axial vibration
- (4) Calculate and plot the corresponding axial vibration modeshapes
- (5) Graduate Work: Research the topic of the split Hopkinson pressure bar and write a short description of this apparatus and its uses

C.9 Analyze Axial Vibration In A Free-Free Bar

Consider a slender bar of finite length L as shown in Figure 11. The bar has cross-section area A , density ρ , elastic modulus E . The bar is free at both ends.

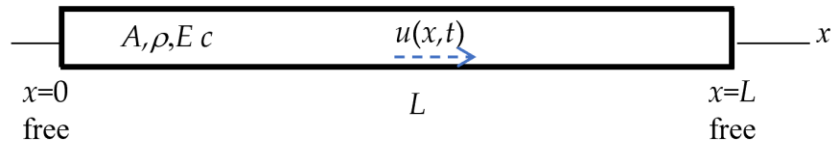


Figure 11 Vibration of a slender bar of finite length L free at both ends

Do the following:

- (1) write the boundary conditions at the two ends of the bar
- (2) recall from Problem C.6 the general solution $u(x, t)$ in terms of:
 - (a) first term: a function $U(x)$ depending on constants C_1, C_2 and trigonometric functions of γx
 - (b) second term: a complex exponential function of ωt
- (3) substitute (2) into (1) and write boundary conditions in terms of constants C_1, C_2
- (4) solve (3) for C_1 and obtain an equation for C_2
- (5) discuss the trivial solution
- (6) choose to search for a nontrivial solution and define the equation that needs to be solved to find a nontrivial solution
- (7) introduce the notation $z = \gamma L$ and write the eigenvalue equation in terms of z
- (8) solve the eigenvalue equation and write the general expression for the eigenvalues z_j , $j = 1, 2, 3, \dots$
- (9) use (8) to write the general expression for the wavenumber γ_j in terms of j and length L
- (10) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c
- (11) use (10) to solve for frequency ω in terms of wavenumber γ and wavespeed c
- (12) use (9) and (11) to write the general expression of the natural frequency ω_j in terms of j , wavespeed c , and bar length L
- (13) use (12) to write the general expression of the Hz frequency f_j in terms of j , wavespeed c , and length L
- (14) explain the meaning of fundamental frequency and overtones
- (15) recall from HW01 the relationship between wavelength λ and wavenumber γ
- (16) use (9) into (15) to write the general expression for the wavelength λ_j in terms of j and length L
- (17) use (16) to explain which wavelength multiples could be accommodated vibratory standing waves in a bar of length L fixed at both ends

- (18) recall $U(x)$ from (2)(a) and use (4) to eliminate C_1
- (19) set $C_2 = 1$ and write the general expression of the modeshape $U_j(x)$ in terms of the wavenumber γ_j
- (20) give the expression of modeshape orthonormality
- (21) write the general expression of the stress modeshapes

C.10 Analyze Axial Vibration In A Fixed-Fixed Bar

Do the following:

- (1) sketch a bar fixed at both ends indicating all the information needed for the analysis
- (2) write the boundary conditions at the two ends of the bar
- (3) repeat the steps of Problem C.9

C.11 Analyze Axial Vibration In A Fixed-Free Bar

Do the following:

- (1) sketch a bar fixed at one end and free at the other end indicating all the information needed for the analysis
- (2) write the boundary conditions at the two ends of the bar
- (3) repeat the steps of Problem C.9